MILITARY SPECIFICATION

CLOTH, CORDED, POLYESTER-COTTON WARP AND POLYESTER FILLING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 *This specification covers cloth constructed of polyester and cotton* warp yarns and polyester filling yarns in the following types:

Type I - Cotton ground yarn
Polyester cord yarn

Type II - Polyester/cotton ground yarn
Polyester/cotton cord yarn

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-P-1134 - *Packaging and Packing of Cotton and Cotton-Synthetic* Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

FSC 8305

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATION

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. The cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Cotton. The cotton shall be carded and combed.

3.2.2 Polyester fiber. The fiber shall be polyethylene glycol terephthalate.

3.2.3 Yarns.

3.2.3.1 Type I.

3.2.3.1.1 Ground warp yarns. The ground warp yarns shall be singles combed cotton yarn.

3.2.3.1.2 Cord warp yarns. For Shade Blue 3304, the cord warp yarns shall be spun from superbrightened polyester fiber. For Shade AG-160, the cord warp yarns may be spun from either semi-dull or superbrightened polyester fiber. The cord warp yarns for both shades shall be 30/3 $\pm$ 10 percent (cotton system) using a minimum of 75 percent nominal 3 denier staple and the remaining percent nominal 1 1/2 denier staple.

3.2.3.1.3 Filling yarns. For Shade Blue 3304, superbrightened polyester fiber shall be used. For Shade AG-160, either semi-dull or superbrightened fiber may be used. The filling yarns for both shades shall be nominal 150 denier multifilament or nominal 70 denier 2 ply multifilament.

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* 3.2.3.2 Type II.

3.2.3.2.1 Ground warp yarns. The ground warp yarns shall be a blend of 65 ± 3 percent semi-dull, nominal 1 1/2 denier polyester and the remaining percentage cotton.

3.2.3.2.2 Cord warp yarns. The cord warp yarns shall be a blend of 65 ± 3 percent semi-dull, nominal 1 1/2 denier polyester and the remaining percentage cotton.

3.2.3.2.3 Filling yarns. The filling yarns shall be either semi-dull or superbrightened polyester fiber, nominal 150 denier multifilament or nominal 70 denier 2 ply multifilament.

* 3.3 Color. The color of the cloth shall conform to a standard shade sample (see 6.2). For Shade Blue 3304, the color shall be obtained by weaving dyed cotton yarns and white superbrightened polyester yarns (type I), or by weaving dyed semi-dull polyester and cotton ground warp yarns and undyed semi-dull polyester and cotton cord yarns (type II). For Shade AG-160, the color shall be obtained by weaving dyed cotton yarns and either semi-dull or superbrightened polyester yarns (type I), or be weaving dyed semi-dull polyester and cotton ground warp yarns and undyed semi-dull polyester and cotton cord yarns (type II). When superbrightened polyester yarns are used for Shade AG-160, the brightness of the shade shall be toned down by the application of dyes (see 6.3.1) either to the polyester yarn prior to weaving or to the woven cloth to match the standard shade sample.

* 3.3.1 Dyeing. The color of the cotton (type I), or polyester-cotton (type II) ground warp yarns shall be obtained by yarn dyeing. The polyester fiber in the warp and filling shall be the color as received from the fiber producer. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuffs for the cotton component shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.3. When a standard sample is not available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.3.

3.3.2 Matching. The color of the finished cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin, and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.3 Colorfastness. The finished cloth shall show colorfastness to laundering, light, perspiration and crocking equal to or better than the standard sample. When no standard sample is available, the finished cloth shall show "good" colorfastness to laundering, light and perspiration and a Munsell value for crocking not lower than 8.5, when tested as specified in 4.3.

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* 3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.3.

TABLE I. Physical requirements

Weight/square yard (minimum)	Yarns per inch (minimum)			Breaking strength in pounds (minimum)	
Ounces	Ground warp	Cord warp	Filling	Warp 1/	Filling
				Pounds	Pounds
4.3	56	28	60	90	110

1/ The initial break obtained (i.e., the breaking of the ground ends) shall be considered the warp breaking strength of the cloth.

3.4.1 Width. Unless otherwise specified (see 6.2), the finished cloth shall be a minimum of 45 inches inclusive of selvages.

3.4.2 Weave. The weave shall be plain. The warp stripe effect shall be produced by alternating 2 ends of ground yarn and one end of cord yarn.

3.5 Finish.

3.5.1 Nonfibrous material content. The starch and protein content (including chloroform-soluble and water-soluble material) of the finished cloth shall be no more than 2.0 percent when tested as specified in 4.3.

3.6 pH. The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.3.

3.7 Dimensional stability. The finished cloth shall not shrink or elongate more than 2.0 percent in either the direction of the warp or of the filling when tested as specified in 4.3. The preshrinking process used shall not be identified by name or trademark, either on the cloth, ticket, or package.

3.8 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up on rolls as specified in PPP-P-1134.

3.9 Fiber identification. Each roll of cloth shall be labeled, ticketed or invoiced for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.10 Marking. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the piece.

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3.11 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.2.2 Examination of the end item. Defects found during examination shall be classified in accordance with 4.2.2.1 thru 4.2.2.4.

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. Defects as defined in Section III of FED-STD-4 which are clearly visible at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scales (see 6.4), E for slubs and D for knots, shall be scored. No linear yard or increment of one yard on the measuring device of the inspection machine from any one roll within the sample, shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. For determination of acceptability on the average quality of the lot, the lot shall be unacceptable if the points per 100 square yards

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exceed 25 points. For determination of acceptability on the individual roll quality basis, the lot shall be unacceptable if two or more individual rolls exceed 38 points. If one roll exceeds the point level, a second sample of 20 rolls shall be selected for individual roll quality examination only. If one or more rolls exceed the value, the lot shall be rejected. Point computation for the lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth
Width of cloth less than specified
Poor dye penetration, mottled, streaky or cloudy
Excessive neppiness

4.2.2.2 Examination for length.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the ticket.

4.2.2.3 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

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4.2.2.4 Examination for identification of preshrinkage process; noncompliance with the Rules and Regulations Under the Textile Fiber Products Identification Act; and face markings. During the yard-by-yard examination each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act, or the face stamping missing from either, or both ends, or the face stamping on the wrong side.

4.2.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking comply with the section 5 requirements.

4.3 Testing of the end item. The methods of testing specified in FED-STD-191 wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be 3-1/2 continuous yards full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The lot size shall be expressed in units of 1 yard each. The sample size (number of sample units) shall be in accordance with the following:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Test methods

Characteristic	Requirement paragraph	Test method
Cotton identification	3.2.1	1200 1/
Polyester identification	3.2.2	4.4.1 1/
Ground warp yarn:		
Singles cotton (type I)	3.2.3.1.1	1/
Polyester/cotton content (type II)	3.2.3.2.1	4.4.2 1/
Polyester denier (type II)	3.2.3.2.1	4021 1/
Yarn dyed (types I and II)	3.3.1	1/

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TABLE II. Test methods (cont'd)

Characteristics	Requirement paragraph	Test method
Cord warp yarn:		
Polyester (type I)	3.2.3.1.2	1/
Polyester/cotton (type II)	3.2.3.2.2	4.4.2 1/
Denier (types I and II)	3.2.3.1.2 and 3.2.3.2.2	4021 1/
Filling yarn (types I and II):		
Polyester	3.2.3.1.3 and 3.2.3.2.3	1/
Denier	3.2.3.1.3 and 3.2.3.2.3	4021 1/
Labile sulfur	3.3.1	2020
Weight, per square yard	3.4	5041
Yarns per inch (Warp and Filling)	3.4	5050
Breaking strength (Warp and Filling)	3.4	5100
Colorfastness to:		
Laundering	3.3.3	5614 3/
Light	3.3.3	5660
Perspiration	3.3.3	5680
Crocking	3.3.3	5651
Weave	3.4.2	Visual 2/
Nonfibrous materials	3.5.1	2611 1/
pH	3.6	2811
Dimensional stability (Warp and Filling)	3.7	5552

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ One determination shall be made from each sample unit and the result shall be reported as "pass" or "fail".

3/ On the color transfer cloth evaluation, only the stain on the nylon and cotton fibers of the color transfer cloth shall be evaluated.

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4.4 Tests.

4.4.1 Polyester fiber. The polyester filament shall be insoluble in each of the following solvents: acetone (boiling); 70 percent sulfuric acid; 5.0 percent sodium hydroxide (boiling); and methylene chloride (boiling). The methylene chloride shall be prepared by mixing nine parts reagent methylene chloride with one part U. S. P. (95 percent) ethyl alcohol. The methylene chloride solution is extremely toxic and adequate ventilation or use of fume hood is recommended. Unless otherwise specified, the above solvents shall be used at room temperature, using 100 times as much reagent as fiber. The fiber shall remain in the solvent for ten minutes with intermittent stirring. This is a visual qualitative test.

4.4.2 Test for polyester fiber and cotton content. The general procedures of Method 2100 shall be followed and the fiber content percentages shall be calculated as follows:

$$\frac{\text{Weight of dry residual fiber} \times 100}{\text{Weight of dry desized specimen}} = \text{percent polyester}$$

100 - percent polyester = percent cotton

4.4.2.1 Report. Two specimens shall be tested from each sample unit and the average percent polyester and the average percent cotton of the two specimens shall be reported to the nearest 1.0 percent.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in women's summer uniforms.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Color of cloth required (see 3.3).
(c) Width of cloth required when other than specified (see 3.4.1).
(d) When length is other than specified (see 3.8).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.3.1 Shade standard, Army. The shade standard submitted for AG 160 has been obtained by dyeing the cotton yarn with the following formulation:

Vat Green No. 3, Color Index No. 69500
Vat Green No. 1, Color Index No. 59825
Vat Blue No. 4, Color Index No. 69800

A successful procedure for toning down this shade made from superbrightened polyester yarns has been as follows:

Pad and reduce by conventional continuous methods a dye formulation consisting of Vat Green No. 8, CI 71050. Oxidize with sodium perborate or hydrogen peroxide and after soap. Finish in the conventional manner.

* 6.4 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 769, "Fabric Defect Replica Kit", 925 South Homan Avenue, Chicago, Illinois 60607.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
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DOCUMENT IDENTIFIER AND TITLE

MIL-C-40052E CLOTH, CORDED, POLYESTER-COTTON WARP AND POLYESTER FILLING

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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